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# Thermodynamic Assessments of the Sc-M (M: Cr, Gd, Mo, W and Zr) Systems

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**By using the calculation of phase diagrams (CALPHAD) method, the thermodynamic assessments of the Sc-M (M: Cr, Gd, Mo, W and Zr) binary systems were carried out based on the available experimental data including thermodynamic properties and phase equilibria. Gibbs free energies of the liquid, hcp and bcc phases in the Sc-M (M: Cr, Gd, Mo, W and Zr) binary systems were modeled by the substitutional solution model. An agreement between the present calculated results and experimental data was obtained.**

**Keywords** CALPHAD, phase diagram, thermodynamic assessment, thermodynamic properties

## 1. Introduction

Sc-based alloys have been used as aero-structures materials,[1] and Sc addition on wrought alloys has positive effects on weldability and welding properties.[2] The phase equilibria in the Sc-based alloy systems are of importance for design of new Sc-based alloys. The CALPHAD method, which is a powerful tool to decrease cost and time during development of materials,[3] effectively provides a clear guideline for material design. As a result, it is of great essence to establish the thermodynamic database for the Sc-base alloy systems. As part of this thermodynamic database, thermodynamic assessments of Sc-M (M: Cr, Gd, Mo, W and Zr) binary systems become necessary.

Recently, the Sc-Cr system was thermodynamically assessed by Bu et al.[4] without any thermodynamic data. In the present optimization, the predicted enthalpy of mixing of liquid phase[5] is employed. Lele and Sarma[6] optimised the Sc-Zr system by means of cluster variation method. However, the liquid phase was not considered.[6] Later, Du et al.[7] predictied precipitation kinetics for Al-0.09 at.% Sc-0.03 at.% Zr alloy with a mathematical model coupled to CALPHAD.

In the present paper, the thermodynamic descriptions for the phase diagrams in the Sc-M (M: Cr, Gd, Mo, W and Zr) binary systems were carried out by combining thermodynamic models with the experimental data in the framework of the CALPHAD method.

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## 2. Experimental Information

There are no intermetallic compounds reported in the Sc-M (M: Cr, Gd, Mo, W and Zr) binary systems. De Boer and Pettifor[5] predicted the value of the enthalpy of mixing of liquid phase at 50 at.% Sc in these five binary systems is 1 kJ/mol (Sc-Cr system), 1 kJ/mol (Sc-Gd system), 11 kJ/mol (Sc-Mo system), 9 kJ/mol (Sc-W system) and 4 kJ/mol (Sc-Zr system), respectively; however, no temperature dependence was given in the work of de Boer and Pettifor.[5]

### 2.1 The Sc-Cr System

The Sc-Cr system is consisted of three solid solution phases ((Cr), αSc and βSc phases). Based on differential thermal analysis (DTA), microhardness measurements, x-ray diffraction (XRD) and metallographic methods, the Sc-Cr system was investigated by Svechnikov et al.,[8] who determined a eutectic reaction (L (~65 at.% Sc) ↔ (Cr) + βSc at 1090 °C) and a eutectoid reaction (βSc ↔ (Cr) + αSc at 990 °C). Svechnikov et al.[8] also reported the solubility of Cr in the (βSc) phase must be less than 10 at.% and the maximum solubility of Sc in the (Cr) phase should be less than 2 at.% at 1090 °C. However, Geiselman[9] reported that the solubility of Sc in the (Cr) phase is to be less than 0.35 at.% by using the metallography, and Tayler[10] considered it to be less than 0.1 at.% by means of XRD. Later, The phase diagram of the Sc-Cr system was reviewed by Venkatraman and Neumann.[11]

### 2.2 The Sc-Gd System

In the Sc-Gd system, there are two reciprocal solid solution phases (αSc and αGd, βSc and βGd phases). Based on thermal analysis, optical pyrometer and x-ray diffraction methods, the phase diagram in the Sc-Gd system was measured by Beaudry and Daane,[12] who indicated the minimum solidus at 35 at.% Sc and 1275 °C, and the minimum transition temperature from the hcp phase to the bcc phase at 45 at.% Sc and 1125 °C. But, the melting temperatures and the β ↔ α transformation temperatures did

not agree with the values of the pure metals in Ref 13. Consequently, the Sc-Gd binary phase diagram was compiled by Gschneidner and Calderwood[14] with some modifications from the experimental data reported by Beaudry and Daane.[12]

### 2.3 The Sc-Mo System

The Sc-Mo phase diagram consisted of three solid solution phases ((Mo), αSc and βSc phases) is mainly based on the work of Massalski,[15] who indicated a eutectic reaction (L (~89.5 at.% Sc) ↔ (Mo) (~0.2 at.% Sc) + βSc (~98.5 at.% Sc) at 1370 ± 50 °C) and a eutectoid reaction (βSc (~98.5 at.% Sc) ↔ (Cr) (~0.2 at.% Sc) + αSc (99.5 at.% Sc) at 1330 °C).

### 2.4 The Sc-W System

The solid solution phases of the Sc-W system are W(bcc), αSc(hcp) and βSc(bcc) above room temperature. Firstly, by using the optical microscope, x-ray diffraction, and mass spectrometry, Taylor et al.[16] determined the solubility of Sc in (W) to be negligible. By means of equilibration and quenching method, Dennison et al.[17] reported some solubility data of W in liquid Sc and indicated a eutectic reaction: L (98.67 at.% Sc) ↔ (W) + βSc at 1510 °C. The liquidus curve was simply extrapolated to higher rangers of compositions by Pandian et al.,[18] who suggested the eutectic composition is to be 98.0 at.% Sc. Later, Okamoto[19] reviewed the phase diagram of the Sc-W system.

### 2.5 The Sc-Zr System

The phase diagram of the Sc-Zr system exhibits two reciprocal solid solution phases (αSc and αZr, βSc and βZr phases). Based on thermal analysis, optical pyrometer and x-ray diffraction methods, the phase diagram in the Sc-Zr system was established by Beaudry and Daane,[20] who indicated the maximum transition temperature from the hcp phase to the bcc phase at 70 at.% Sc is 1415 °C. Later, The phase diagram of the Sc-Zr system was reviewed by Palenzona and Ciraici.[21]

## 3. Thermodynamic Models

In the Cr-Sc, Gd-Sc, Mo-Sc, Sc-W and Sc-Zr binary systems (A-B system), Gibbs free energies of the liquid, hcp and bcc phases are described by the substitutional solution model, as follows:

$$G_m^\phi = \sum_{i=A,B} {}^0G_i^\phi x_i + RT \sum_{i=A,B} x_i \ln x_i + \Delta^E G_m^\phi, \qquad \text{(Eq 1)}$$

where ${}^0G_i^\phi$ is the Gibbs free energy of the pure component $i$ in the respective reference state with ϕ phase, which is taken from the SGTE pure element database;[22] $x_i$ denotes the mole fraction of the component $i$; $R$ is the gas constant; $T$ is the absolute temperature; and the term $\Delta^E G_m^\phi$ presents the excess Gibbs free energy, which is expressed in the Redlich-Kister polynomial[23] as:

$$\Delta^E G_m^\phi = x_A x_B \sum_{m=0}^{n} {}^mL_{A,B}^\phi (x_A - x_B)^m, \qquad \text{(Eq 2)}$$

$${}^mL_{A,B}^\phi = a + bT, \qquad \text{(Eq 3)}$$

where ${}^mL_{A,B}^\phi$ is the binary interaction parameter, and the coefficients of $a$ and $b$ are evaluated on the basis of available experimental data.

## 4. Optimized Results and Discussion

The optimization of the thermodynamic parameters was carried out by using the PARROT program in the Thermo-Calc software,[24] which can handle various kinds of experimental data. The experimental data of the phase diagram and thermodynamic properties were used as input to the program. Each piece of selected information was given a certain weight based on the importance of data, and changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits.

In the present work, considering the eutectic reaction, eutectoid reaction, solid solubility in the Sc-rich region, and the predicted enthalpy of mixing of liquid phase, the temperature dependent parameters and higher order parameters (${}^1$L) were utilized in the Sc-Cr and Sc-Mo systems. Especially in the Sc-Cr system, the ${}^2$L parameter was employed to describe the liquid phase better. Since the compositions of the minimum solidus (L ↔ (βGd, βSc) at 35 at.% Sc) in the Sc-Gd system and the transition ((βSc, βZr) ↔ (αSc, αZr) at 70 at.% Sc) in the Sc-Zr system are far away from 50 at.% Sc, it was necessary to use the temperature dependent parameters and higher order parameters (${}^1$L and ${}^2$L) in these binary systems.

The calculated compositions and temperatures for the invariant reactions compared with the experimental data in the Sc-M (M: Cr, Gd, Mo, W and Zr) binary systems are list in Table 1. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Sc-M (M: Cr, Gd, Mo, W and Zr) systems are given in Table 2. Table 3 shows the comparison of the calculated and the predicted[5] enthalpies of mixing of the liquid phase at 50 at.% Sc in the Sc-M (M: Cr, Gd, Mo, W and Zr) binary systems.

### 4.1 The Sc-Cr System

The calculated Sc-Cr phase diagram compared with the experimental data is shown in Fig. 1. It is seen that the calculated results are in good agreement with the experimental data.[8–10] The calculated eutectoid reaction (βSc(93.4 at.% Sc) ↔ (Cr) + αSc(98.62 at.% Sc)) occurs at 990.1 °C, and the calculated eutectic reaction (L (65.2 at.% Sc) ↔ (Cr) + βSc (92.3 at.% Sc)) is located at 1090 °C. The solubility of Sc in (Cr) is less than 0.1 at.% Sc at 1090 °C. Figure 2 shows the calculated enthalpies of mixing of the liquid phase at 2000 °C with reference states

**Table 1 Experimental and calculated invariant reactions in the Sc-M (M: Cr, Gd, Mo, W, Zr) binary systems**

| System | Invariant Reaction | Composition at.% Sc | | | T, °C | Reference |
|---|---|---|---|---|---|---|
| Sc-Cr | L ↔ (Cr) + βSc | ~65 | <2.0 | >90 | 1090 | [8] |
| | | | <0.35 | | | [9] |
| | | | <0.1 | | | [10] |
| | | 65.2 | <0.1 | 92.3 | 1090 | This work |
| | βSc ↔ (Cr) + αSc | | | | 990 | [8] |
| | | 93.4 | 0 | 98.62 | 990.1 | This work |
| Sc-Gd | L ↔ (βGd, βSc) | | 35 | | 1275 | [12] |
| | | | 34.4 | | 1275 | This work |
| | (βGd, βSc) ↔ (αGd, αSc) | | 45 | | 1125 | [12] |
| | | | 45.6 | | 1125 | This work |
| Sc-Mo | L ↔ (Mo) + βSc | ~89.5 | ~0.2 | ~98.5 | 1370 ± 50 | [15] |
| | | 89.6 | 0.2 | 98.3 | 1370 | This work |
| | βSc ↔ (Mo) + αSc | ~98.5 | ~0.2 | ~99.5 | 1330 | [15] |
| | | 98.5 | 0.2 | 99.8 | 1329.9 | This work |
| Sc-W | L ↔ βSc + (W) | 98.67 | 0 | 100 | 1510 | [17] |
| | | 98 | 0 | 100 | 1510 | [18] |
| | | 98.35 | 0 | 100 | 1510.1 | This work |
| Sc-Zr | (βSc, βZr) ↔ (αSc, αZr) | | 70 | | 1415 | [20] |
| | | | 70.4 | | 1415 | This work |

**Table 2 Thermodynamic parameters in the Sc-M (M: Cr, Gd, Mo, W and Zr) binary systems assessed in the present work**

| System | Phase | Parameters in each phase, J/mol |
|---|---|---|
| Sc-Cr | Liquid phase | ${}^{0}L^{Liq}_{Cr,Sc} = 3900 + 2.4T$ |
| | | ${}^{1}L^{Liq}_{Cr,Sc} = 4.8T$ |
| | | ${}^{2}L^{Liq}_{Cr,Sc} = 1500 + 4.1T$ |
| | Bcc phase | ${}^{0}L^{Bcc}_{Cr,Sc} = 47900 + 3T$ |
| | | ${}^{1}L^{Bcc}_{Cr,Sc} = 25800$ |
| | Hcp phase | ${}^{0}L^{Hcp}_{Cr,Sc} = 41650$ |
| Gd-Sc | Liquid phase | ${}^{0}L^{Liq}_{Gd,Sc} = 3970 + 3.05T$ |
| | | ${}^{2}L^{Liq}_{Gd,Sc} = 3230$ |
| | Bcc phase | ${}^{0}L^{Bcc}_{Gd,Sc} = -169 + 8.413T$ |
| | | ${}^{1}L^{Bcc}_{Gd,Sc} = -756$ |
| | | ${}^{2}L^{Bcc}_{Gd,Sc} = 2149$ |
| | Hcp phase | ${}^{0}L^{Hcp}_{Gd,Sc} = 237 + 9.12T$ |
| | | ${}^{1}L^{Hcp}_{Gd,Sc} = -704$ |
| | | ${}^{2}L^{Hcp}_{Gd,Sc} = -1986 + 2.415T$ |
| Sc-Mo | Liquid phase | ${}^{0}L^{Liq}_{Mo,Sc} = 43420 - 12.92T$ |
| | | ${}^{1}L^{Liq}_{Mo,Sc} = 10000$ |
| | Bcc phase | ${}^{0}L^{Bcc}_{Mo,Sc} = 70000$ |
| | | ${}^{1}L^{Bcc}_{Mo,Sc} = 13000$ |
| | Hcp phase | ${}^{0}L^{Hcp}_{Mo,Sc} = 46500$ |
| Sc-W | Liquid phase | ${}^{0}L^{Liq}_{Sc,W} = 35000$ |
| | Bcc phase | ${}^{0}L^{Bcc}_{Sc,W} = 140000$ |
| | Hcp phase | ${}^{0}L^{Hcp}_{Sc,W} = 79000$ |
| Sc-Zr | Liquid phase | ${}^{0}L^{Liq}_{Sc,Zr} = 16000 - 0.8T$ |
| | Bcc phase | ${}^{0}L^{Bcc}_{Sc,Zr} = 2550 + 6.7T$ |
| | Hcp phase | ${}^{0}L^{Hcp}_{Sc,Zr} = 1650 + 5.71T$ |
| | | ${}^{1}L^{Hcp}_{Sc,Zr} = 923$ |
| | | ${}^{2}L^{Hcp}_{Sc,Zr} = -1130$ |

**Table 3 Comparison of the calculated and the predicted[5] enthalpies (kJ/mol) of mixing of the liquid phase at 50 at.% Sc in the Sc-M (M: Cr, Gd, Mo, W and Zr) binary systems**

| System | Predicted[5] | Calculated |
|---|---|---|
| Sc-Cr | 1 | 0.975 |
| Sc-Gd | 1 | 0.993 |
| Sc-Mo | 11 | 10.86 |
| Sc-W | 9 | 8.75 |
| Sc-Zr | 4 | 4 |

The reference states of pure elements of Sc and M (M: Cr, Gd, Mo, W and Zr) are liquid phases

of pure Sc (liquid) and Cr (liquid) in the Sc-Cr binary system. The calculated value at 50 at.% Sc is 0.975 kJ/mol, which is very close to the predicted values of 1 kJ/mol.[5]

### 4.2 The Sc-Gd System

The optimized phase diagram of the Sc-Gd system in relation to the experimental data from the literature is shown in Fig. 3. The calculated Sc-Gd phase diagram agrees well with the experimental data[12] to an extent. The calculated enthalpies of mixing of the liquid phase at 1600 °C with reference states of pure Sc (liquid) and Gd (liquid) is shown in Fig. 4, and the calculated value at 50 at.% Sc is 0.993 kJ/mol, which shows good agreement with the predicted values of 1 kJ/mol.[5]

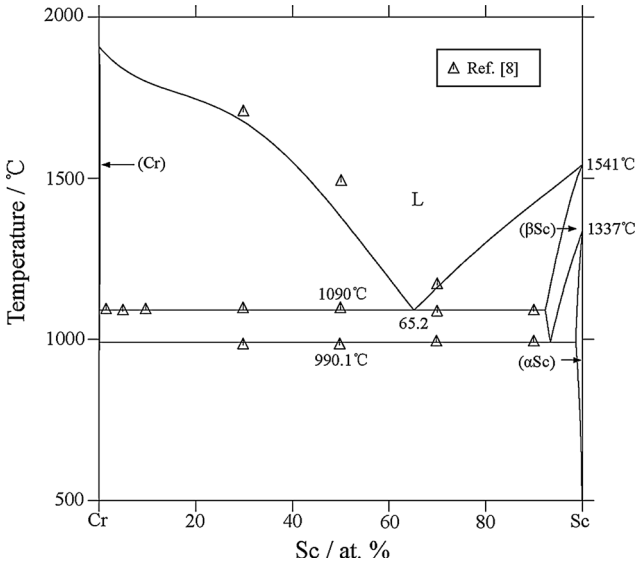


**Fig. 1** Calculated phase diagram of the Sc-Cr system compared with the experimental data[8]

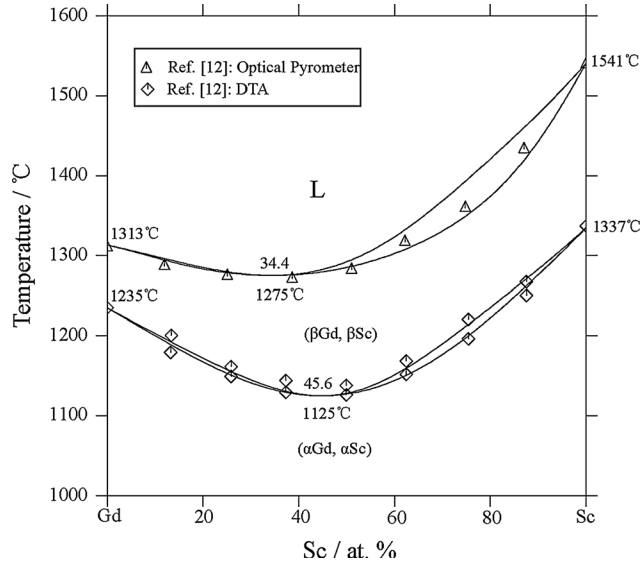


**Fig. 3** Calculated phase diagram of the Sc-Gd system compared with the experimental data[12]

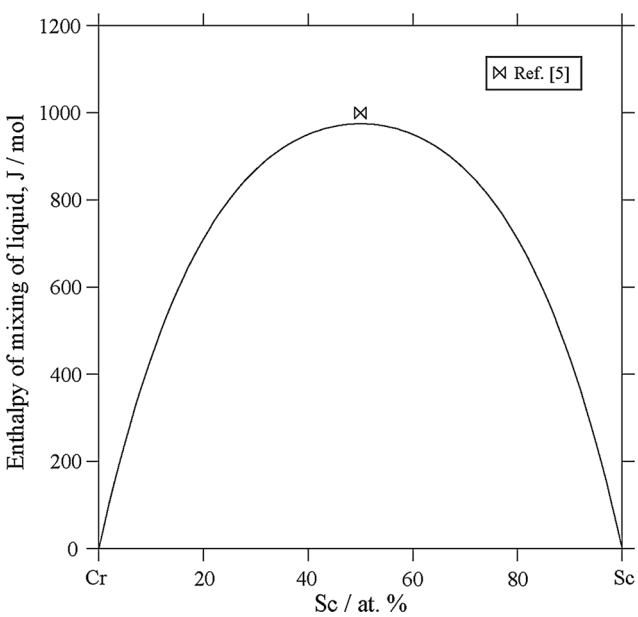


**Fig. 2** Calculated enthalpies of mixing of liquid phase at 2000 °C with reference states of pure elements of Sc (liquid) and Cr (liquid) with the predicted data[5]

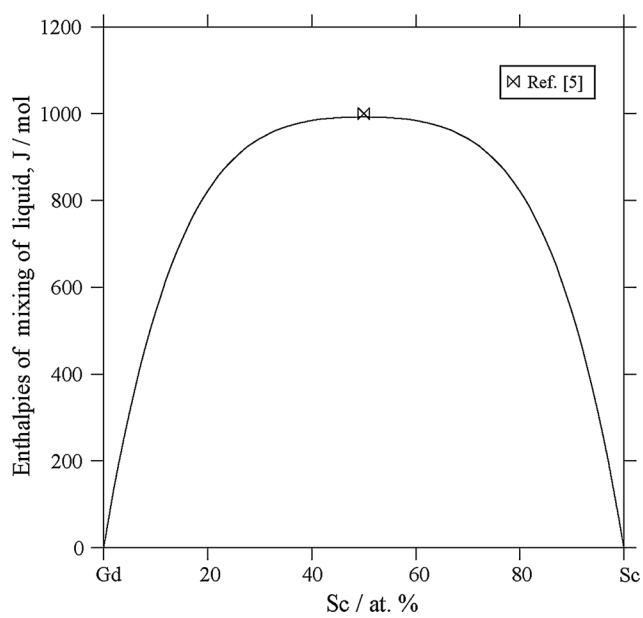


**Fig. 4** Calculated enthalpies of mixing of liquid phase at 1600 °C with reference states of pure elements of Sc (liquid) and Gd (liquid) with the predicted data[5]

### *4.3 The Sc-Mo System*

The calculated phase diagram of the Sc-Mo binary system is presented in Fig. 5. The assessed phase diagram by Massalski[15] can be well reproduced by the present calculation. Figure 6 shows the calculated enthalpies of mixing of the liquid phase at 2700 °C with reference states of pure Sc (liquid) and Mo (liquid) in the Sc-Mo binary system. As can be seen in this figure, the calculated value (10.86 kJ/mol) at 50 at.% Sc is very close to the predicted values (11 kJ/mol).[5]

Based on the optimized parameters, the extrapolated miscibility gap in the liquid phase was also predicted in the Sc-Mo system, as shown in Fig. 5 by dotted lines. The calculated critical temperature and composition of the liquid miscibility gap are 1421.2 °C and 30.7 at.% Sc in the present work.

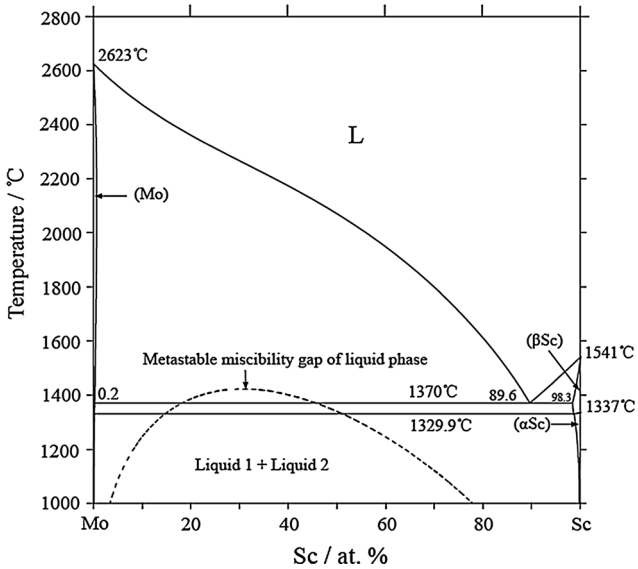


**Fig. 5** Calculated phase diagram of the Sc-Mo system

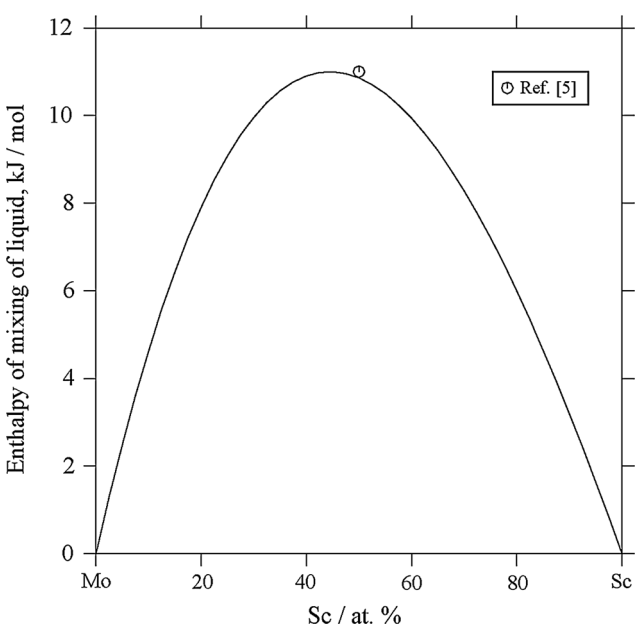


**Fig. 6** Calculated enthalpies of mixing of liquid phase at 2700 °C with reference states of pure elements of Sc (liquid) and Mo (liquid) with the predicted data[5]

### 4.4 The Sc-W System

Figure 7 presents the optimized Sc-W phase diagram compared with the experimental data. According to the calculation, the temperature and composition of the eutectic reaction (L ↔ βSc + (W)) are 1510.1 °C and 98.35 at.% Sc, which are the same as the experimental data[17] basically. Figure 8 shows the calculated enthalpies of mixing of the liquid phase at 3500 °C with reference states of pure Sc (liquid) and W (liquid). The calculated value at

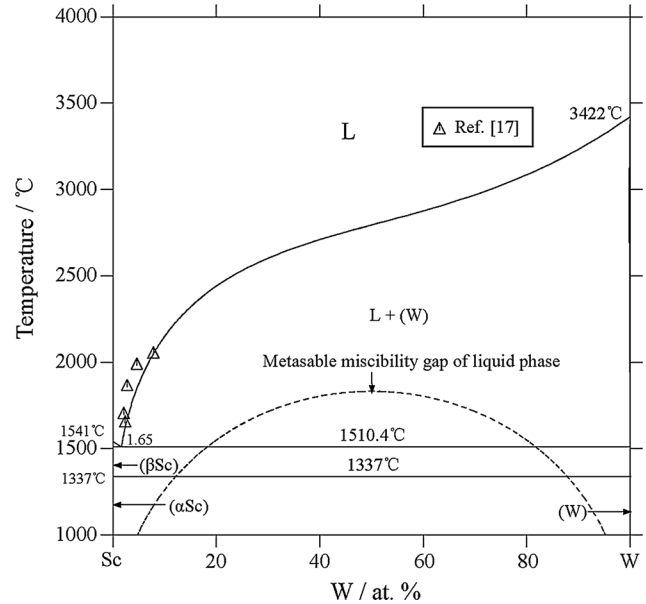


**Fig. 7** Calculated phase diagram of the Sc-W system compared with the experimental data[17]

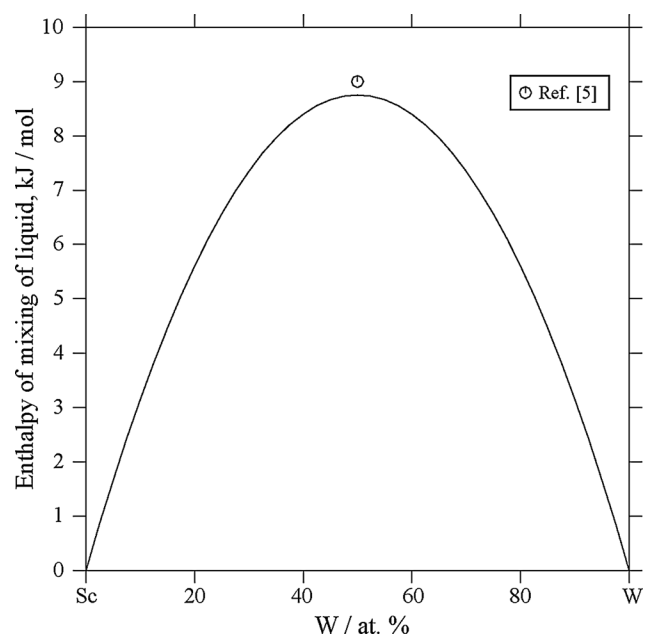


**Fig. 8** Calculated enthalpies of mixing of liquid phase at 3500 °C with reference states of pure elements of Sc (liquid) and W (liquid) with the predicted data[5]

50 at.% Sc is 8.75 kJ/mol, which is close to the predicted values of 9 kJ/mol.[5]

According to the present optimized thermodynamic parameters, the calculated metastable miscibility gap of the liquid phase in the Sc-W system can be observed in Fig. 7. The calculated critical temperature and composition

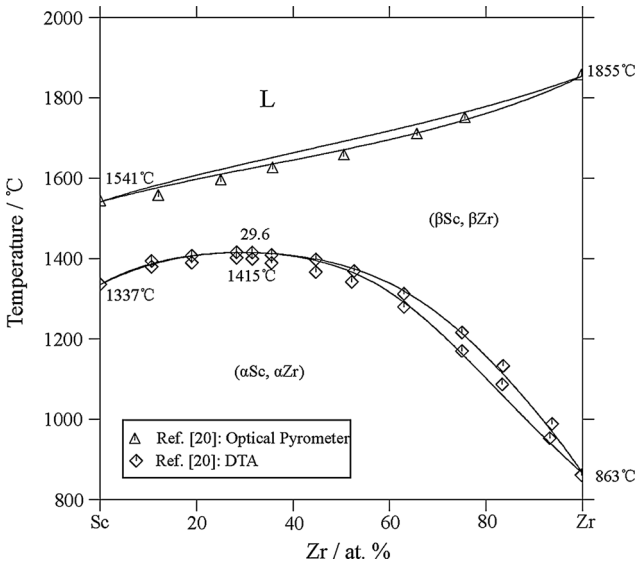


**Fig. 9** Calculated phase diagram of the Sc-Zr system compared with the experimental data[20]

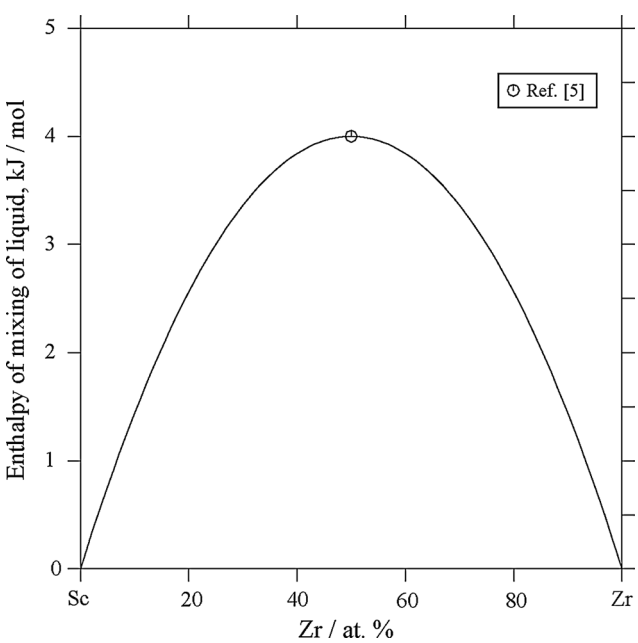


**Fig. 10** Calculated enthalpies of mixing of liquid phase at 1900 °C with reference states of pure elements of Sc (liquid) and Zr (liquid) with the predicted data[5]

of the metastable miscibility gap of the liquid phase are 1831.5 °C and 50 at.% W. More thermodynamic data in Sc-W binary system need to be determined, and the thermodynamic assessment work should be carried out with new thermodynamic data in the future.

### *4.5 The Sc-Zr System*

The comparison between the calculated Sc-Zr phase diagram and the experimental data[20] is shown in Fig. 9, and the calculated phase diagram fit the experimental data well. Also, good agreement between the calculated and the predicted[5] enthalpies of mixing of the liquid phase at 1900 °C at 50 at.% Sc with reference states of pure Sc (liquid) and Zr (liquid) can be observed in Fig. 10.

## 5. Conclusions

The phase diagram in the Sc-M (M: Cr, Gd, Mo, W and Zr) binary systems were thermodynamically assessed based on the experimental data including phase equilibria and thermodynamic data, and most of the experimental information can be satisfactorily reproduced on the basis of the optimized thermodynamic parameters.


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